

D888

Probability and Statistics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A middle school math teacher who runs the chess club surveys students about after-school activities. 40% play a sport, 30% are in music, and 12% do both. The survey was done online during homeroom. What is the probability that a randomly chosen student plays a sport or is in music?

- A. 0.70
- B. 0.12
- C. 0.82
- D. 0.58

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For two events, $P(A \text{ or } B) = 0.70$, $P(A) = 0.50$, and $P(B) = 0.40$. What is $P(A \text{ and } B)$?

- A. 0.10
- B. 0.20
- C. 0, because the events are mutually exclusive.
- D. 0.90

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher builds a two-way table from a school survey that took place during spirit week. Of 200 students, 120 play a sport, 80 are in band, and 45 do both. The table will be printed for a lesson on data. What is the probability that a student is in band, given that the student plays a sport?

- A. 0.375
- B. 0.225
- C. 0.40
- D. 0.5625

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

The student council at a school with a new gym has 10 members. The advisor needs to choose 3 members to form a planning committee with no special roles. The committee meets on Wednesdays. How many different committees are possible?

- A. 30
- B. 720
- C. 120
- D. 1,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

From 10 members, a club must pick a president, a vice president, and a secretary. How many ways are there?

- A. 120
- B. 1,000
- C. 27
- D. 720

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A district wants to survey its high school students about homework. The survey was designed by a committee of teachers over the summer. It randomly selects 50 students from each of grades 9, 10, 11, and 12. What sampling method is this?

- A. Cluster sampling
- B. Systematic sampling
- C. Stratified random sampling
- D. Convenience sampling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A poll of 100 families at a district open house has a margin of error of about 6 percentage points. The open house had free snacks and a raffle. If the district surveys 400 families instead, about what will the margin of error be?

- A. About 3 percentage points
- B. About 1.5 percentage points
- C. About 6 percentage points
- D. About 12 percentage points

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A survey of 250 people has a margin of error of 5 points. About how many people are needed for a margin of 2.5 points?

- A. 625
- B. 1,000
- C. 125
- D. 500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

From a random sample of 60 eighth graders, a teacher finds a 95% confidence interval for mean reading time of 62 to 70 minutes per week. The sample was drawn from the school's roster by software the district licenses. Which interpretation is correct?

- A. 95% of the sample means will be exactly 66.
- B. 95% of eighth graders read between 62 and 70 minutes per week.
- C. The true mean is always between 62 and 70.
- D. The method used gives intervals that capture the true mean about 95% of the time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher tests whether a new warm-up routine raises quiz scores. The routine was shared at a summer workshop. With $\alpha = 0.05$, the test gives $p = 0.03$. What is the correct conclusion?

- A.** The routine is proven to work for every class.
- B.** Reject the null hypothesis; the data give evidence that the routine raises scores.
- C.** Fail to reject the null hypothesis.
- D.** There is a 3% chance the null hypothesis is true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A school tests a new math app. The null hypothesis is that it has no effect. The app has bright graphics and a free trial. The school concludes the app works, but in truth it has no effect. What happened?

- A.** No error, because the school made a decision.
- B.** A sampling bias error
- C.** A Type I error
- D.** A Type II error

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A study fails to find that a new tutoring program helps, but in truth the program does help. What error was made?

- A.** A Type II error
- B.** No error, because the results were not significant.
- C.** A Type I error
- D.** A confidence interval error

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A middle school math teacher ~~who runs the chess club~~ surveys students about after-school activities. 40% play a sport, 30% are in music, and 12% do both. ~~The survey was done online during homeroom.~~ What is the probability that a randomly chosen student plays a sport or is in music?

Target word: P(A or B) **Decides:** Overlap of 12% must be subtracted once.

A	Dropped element	Adds without subtracting the overlap.
B	Confusion set	That is the probability of both, not either.
C	Inverted operation	Adds the overlap instead of subtracting it.
D	Correct	$0.40 + 0.30 - 0.12 = 0.58$.

2. Answer B Concept 1 • transfer

For two events, $P(A \text{ or } B) = 0.70$, $P(A) = 0.50$, and $P(B) = 0.40$. What is $P(A \text{ and } B)$?

Target word: solve for the overlap **Decides:** Rearrange the addition rule.

A	Dropped element	Subtracts P(B) from P(A) and ignores the or probability.
B	Correct	$0.50 + 0.40 - 0.70 = 0.20$.
C	Absolute language	If the overlap were 0, P(A or B) would be 0.90.
D	Dropped element	Adds A and B without using the or probability.

3. Answer A Concept 2

A teacher builds a two-way table from a school survey ~~that took place during spirit week.~~ Of 200 students, 120 play a sport, 80 are in band, and 45 do both. ~~The table will be printed for a lesson on data.~~ What is the probability that a student is in band, given that the student plays a sport?

Target word: P(band | sport) **Decides:** Denominator is the sport group.

A	Correct	$45 / 120 = 0.375$.
B	Dropped element	$45 / 200$ is P(band and sport), not the conditional.
C	Confusion set	$80 / 200$ is P(band) with no condition.
D	Inverted operation	$45 / 80$ is P(sport band), the reversed condition.

4. Answer C Concept 3

The student council ~~at a school with a new gym~~ has 10 members. The advisor needs to choose 3 members to form a planning committee with no special roles. ~~The committee meets on Wednesdays.~~ How many different committees are possible?

Target word: count unordered groups **Decides:** Order does not matter.

A	Wrong stem number	Multiplies 10×3 instead of counting groups.
B	Confusion set	$10P3$ counts ordered roles, which the committee does not have.
C	Correct	$10C3 = 720 / 6 = 120$.
D	Adjacent category	10^3 allows repeats, but one person cannot fill two seats.

5. Answer D Concept 3 • transfer

From 10 members, a club must pick a president, a vice president, and a secretary. How many ways are there?

Target word: count ordered roles **Decides:** Roles are different, so order matters.

A	Inverted operation	That is the count when roles do not matter.
B	Adjacent category	Allows one person to hold several roles.
C	Wrong stem number	Uses 3 cubed instead of the 10 members.
D	Correct	$10 \times 9 \times 8 = 720$.

6. Answer C Concept 4

A district wants to survey its high school students about homework. ~~The survey was designed by a committee of teachers over the summer.~~ It randomly selects 50 students from each of grades 9, 10, 11, and 12. What sampling method is this?

Target word: name the method **Decides:** Random samples within each subgroup.

A	Confusion set	Cluster sampling picks whole groups and surveys everyone in them.
B	Adjacent category	Systematic takes every kth person from a list.
C	Correct	Groups are formed, then a random sample is taken within each.
D	Adjacent category	Convenience uses whoever is easiest to reach.

7. Answer A Concept 5

A poll of 100 families at a district open house has a margin of error of about 6 percentage points. ~~The open house had free snacks and a raffle.~~ If the district surveys 400 families instead, about what will the margin of error be?

Target word: new margin of error **Decides:** n grows by 4, so margin shrinks by the square root of $4 = 2$.

A	Correct	$6 / \text{square root of } 4 = 3$.
B	Inverted operation	Divides by 4 instead of by the square root of 4.
C	Dropped element	Ignores the effect of sample size.
D	Inverted operation	A bigger sample shrinks the margin, not grows it.

8. Answer B Concept 5 • transfer

A survey of 250 people has a margin of error of 5 points. About how many people are needed for a margin of 2.5 points?

Target word: solve for n **Decides:** Halving the margin needs four times the sample.

A	Wrong stem number	Multiplies by 2.5, the target margin.
B	Correct	$250 \times 4 = 1,000$.
C	Inverted operation	A smaller sample gives a larger margin.
D	Confusion set	Doubling n does not halve the margin.

9. Answer D Concept 6

From a random sample of 60 eighth graders, a teacher finds a 95% confidence interval for mean reading time of 62 to 70 minutes per week. The sample was drawn from the school's roster ~~by software the district licenses.~~ Which interpretation is correct?

Target word: meaning of the interval **Decides:** Confidence refers to the method, and the interval estimates the mean.

A	Adjacent category	Confuses the interval with a fixed point estimate.
B	Confusion set	The interval is about the mean, not individual students.
C	Absolute language	There is about a 5% risk the method missed.
D	Correct	Confidence is a property of the method over repeated samples.

10. Answer B Concept 7

A teacher tests whether a new warm-up routine raises quiz scores. ~~The routine was shared at a summer workshop.~~ With $\alpha = 0.05$, the test gives $p = 0.03$. What is the correct conclusion?

Target word: decision **Decides:** $p < \alpha$.

A	Absolute language	A test gives evidence, not proof.
B	Correct	$0.03 < 0.05$.
C	Inverted operation	That applies when p is greater than α .
D	Confusion set	A p-value is not the probability that the null is true.

11. Answer C Concept 8

A school tests a new math app. The null hypothesis is that it has no effect. ~~The app has bright graphics and a free trial.~~ The school concludes the app works, but in truth it has no effect. What happened?

Target word: error type **Decides:** Rejected a true null.

A	Absolute language	The decision was wrong given the truth.
B	Adjacent category	Bias is a design problem, not a decision error type.
C	Correct	Rejecting a null hypothesis that is true.
D	Inverted operation	Type II is failing to reject a false null.

12. Answer A Concept 8 · transfer

A study fails to find that a new tutoring program helps, but in truth the program does help. What error was made?

Target word: error type, reversed **Decides:** Failed to reject a false null.

A	Correct	Missing a real effect is a false negative.
B	Confusion set	A non-significant result is not proof of no effect; here it missed a real one.
C	Inverted operation	Type I rejects a true null.
D	Adjacent category	Not one of the two decision error types.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Addition rule with overlap	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$; only for mutually exclusive events is the overlap zero.	Adding the two probabilities and counting the overlap twice.	The word or with two events that can happen together.
2	Conditional probability	$P(A B) = P(A \text{ and } B) / P(B)$: the denominator is the group you are told you are in.	Dividing by the whole population, or by the wrong condition.	The words given, among, or of those who.
3	Permutations vs. combinations	Use permutations when order or roles matter (nPr) and combinations when only the group matters ($nCr = nPr / r!$).	Using permutations for a committee with no roles, or combinations for ranked positions.	Distinct roles or rankings vs. an unordered group.
4	Sampling methods	Stratified sampling randomly samples within every subgroup; cluster sampling randomly picks whole groups; systematic takes every kth; convenience takes whoever is easy.	Calling a sample of whole groups stratified, or treating an easy sample as random.	How groups are formed and whether sampling happens within or among them.
5	Sample size and margin of error	Margin of error shrinks with the square root of n: to cut it in half, multiply the sample size by four.	Thinking doubling n halves the margin of error.	A target margin of error and a current sample size.
6	Interpreting a confidence interval	A 95% confidence level means the method captures the true parameter in about 95% of repeated samples; it says nothing about 95% of individual values.	Saying 95% of the data fall inside the interval.	A reported interval for a mean or proportion and a choice of interpretations.
7	P-value and significance	If the p-value is less than alpha, reject the null hypothesis; the p-value is the probability of data this extreme if the null is true, not the probability the null is true.	Reading the p-value as the chance the null hypothesis is true, or as proof of the alternative.	A p-value and an alpha level in the same stem.
8	Type I and Type II errors	A Type I error rejects a true null hypothesis (false positive); a Type II error fails to reject a false null (false negative).	Swapping the two, or tying the error type to whether the result was good news.	A decision made and the true state of the world.

Course scope (public description): Probability and Statistics offers candidates a comprehensive introduction to the fundamental principles of probability theory and statistical analysis, specifically designed for educators and aspiring statisticians. Beginning with the basics, students will explore essential concepts such as probability rules, conditional probability, and counting techniques, learning how to apply these in real-world contexts and effectively communicate them to K-12 students. Students will then learn sampling methods and estimation techniques, equipping them with the knowledge to gather, analyze, and interpret representative data for statistical analysis. Finally, students will focus on hypothesis testing and statistical inference, where they will learn to conduct and interpret various tests, including confidence intervals, using statistical software. Through a blend of interactive simulations, scenario-based challenges, and reflective activities, this course prepares students to apply these statistical tools in educational settings and beyond, fostering data-driven decision-making and effective teaching practices.